

Phillip Lippe

Motivated PhD student fascinated by research in causality-inspired deep learning and generative models, but still curious to increase horizon in other fields as well. Interested in connecting with other researchers.

Gender Male Age 22 years old Nationality German 📍 Amsterdam, Netherlands
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EXPERIENCE

Student Research Assistant

University of Amsterdam / Ahold Delhaize

📅 Jul 2019 – Dec 2019 📍 Amsterdam, Netherlands

- Research in task-oriented dialogue system at ILPS and AIRLab
- Research topic: Diversifying Dialogue Response Generation with Prototype Guided Paraphrasing ([publication](#))

Student Research Intern

Daimler AG (Autonomous driving, Image Understanding)

📅 Dec 2016 – Mar 2017 📍 Stuttgart, Germany
Apr 2018 – Sep 2018

- Research and development in Deep Learning for real-time multi-class object detection in complex urban traffic scenes
- Bachelor Thesis: *Hierarchical Multi-label Object Detection of Rare Classes for Autonomous Driving* ([pdf](#))

Student Research Intern

Mercedes-Benz Research and Development NA

📅 May 2017 – Aug 2017 📍 Sunnyvale, United States

- Research and development of Deep Generative Video models for predicting agent behavior in traffic environments
- Performed agile software development with Scrum
- Experimenting with unsupervised and curiosity-driven reinforcement learning approaches ([report](#))

EDUCATION

PhD, Artificial Intelligence

University of Amsterdam, QUVA lab

📅 Sep 2020 – Present 📍 Amsterdam, Netherlands

- My research focuses on the intersection of causality and deep learning, in particular in the context of generative models.
- Supervisors: dr. Efstratios Gavves and Taco Cohen

Master of Science, Artificial Intelligence

University of Amsterdam

📅 Sep 2018 – Aug 2020 📍 Amsterdam, Netherlands

- Master Thesis: Categorical Normalizing Flows ([publication](#))
- Final GPA: 9.5 (Dutch grading system)

Bachelor of Engineering, Computer Science

Baden-Wuerttemberg Cooperative State University

📅 Oct 2015 – Sep 2018 📍 Stuttgart, Germany

- Cooperative study program with the specialization in IT Automotive, in cooperation with Daimler AG.
- Final GPA: 1.0 (German grading system)

SKILLS

PyTorch

Tensorflow, JAX, Caffe



Python, Java

MATLAB, C, HTML, SQL



AWARDS

- Best Computer Science undergraduate student at the Baden-Wuerttemberg Cooperative State University 2018
- Awarded SchülerUni Scholarship 2012/13 for taking university courses on Computer Science/Mathematics during high school
- Won price for best application of ML and AI, and best integration of health and Android wear 2.0 device at MBRDNA hackathon 2017
- Reached the top-5 at NeurIPS 2020 challenge "Hateful Memes" ([paper](#))

PROJECTS

Deep Reasoning

- Developed a siamese-based network architecture for learning clause selection heuristic in first-order logic theorem prover
- Work presented at LuxLogAI conference 2018

Teaching assistant

- Teaching assistant for the graduate courses Deep Learning, NLP 1, FAIR, IR 1 and Advanced Topics in Computational Semantics at University of Amsterdam (2019-2021)
- Created and taught a series of implementation tutorials for various Deep Learning topics including Transformers, Adversarial Attacks, Normalizing Flows, and more ([website](#))

PUBLICATIONS

Phillip Lippe and Efstratios Gavves: **Categorical Normalizing Flows via Continuous Transformations**. Published at: International Conference on Learning Representations (ICLR 2021), [link](#).

Phillip Lippe, Pengjie Ren, Hinda Haned, Bart Voorn and Maarten de Rijke: **Diversifying Task-oriented Dialogue Response Generation with Prototype Guided Paraphrasing**. arXiv preprint [arXiv:2008.03391](https://arxiv.org/abs/2008.03391), 2020. Under review.

Phillip Lippe, Nithin Holla, Shantanu Chandra, Santhosh Rajamanickam, Georgios Antoniou, Ekaterina Shutova, Helen Yannakoudakis: **A Multimodal Framework for the Detection of Hateful Memes**. arXiv preprint [arXiv:2012.12871](https://arxiv.org/abs/2012.12871), 2020.